Artemy Urodovskikh

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Skills

Languages: Python, SQL, C
Backend: Django, Django REST Framework, Celery, PostgreSQL, PostGIS, ClickHouse, Redis
Infrastructure: Docker, Kubernetes, Terraform, GitLab CI, S3 (Yandex/MinIO), Nginx, Grafana
Domain: GIS, route optimization (TSP/CVRP), telemetry, billing, DBSCAN clustering, EAV/versioning
Other: FPGA (Xilinx Spartan-6), SDR, IrDA, Git, pytest, Agile, team leadership (5 engineers)

Professional Experience

Backend Team Lead — Bolshaya Troika, Moscow

May 2018 – present · 8+ years

Core product: regional-scale waste-management platform (Python/Django, PostgreSQL, ClickHouse) for government clients in Russia and Kazakhstan. Developed the billing engine and the route planning subsystem from scratch; implemented a standalone TSP solver as a reserve for the internal optimization service. Led a team of five engineers within a 40-developer codebase. Substantial involvement from architecture planning through to production. Owned releases and rollbacks for the core platform.

2022–2026

Cloud-hosted waste management system for government clients (Russia, Kazakhstan).

Route planning (core owner). Developed the subsystem from scratch: background task queue for route calculations, synchronous solver for 10,000+ waypoints per district, 500+ per vehicle. Implemented a standalone TSP solver with graceful degradation. In production since early 2022.

Billing engine (core owner). Designed and developed from scratch. Tariff calculation, accruals, invoice generation for 1M+ individual payers (municipal waste collection fees). Memory-optimized batch printing system. Also led the cross-cloud storage migration for the platform. In production since late 2022.

Fleet telemetry (contributor). Developed the data synchronization layer between the analytical database (ClickHouse) and the geospatial database (PostGIS). GPS on stationary waste trucks drifts by tens of metres, producing false trips — designed and implemented filtering heuristics (track splitting, segment merging, outlier rejection) that recover the actual geometry of movement from raw GPS streams. Integrated S3 storage for onboard camera footage.

Statistics and analytics (core owner). Designed the analytical subsystem from domain model to operational dashboards: scheduled ETL processing, geospatial clustering (DBSCAN) for map aggregation, two-stage weighing reconciliation (pickup vs. delivery), ClickHouse for regional-scale analytical queries.

Object versioning (core owner). Designed a parallel versioning strategy for domain objects — multiple concurrent drafts with change-set merging on publish. Delta storage: versions can store only changed fields instead of full copies. Relationship copying logic (one-to-many, many-to-many) configurable through admin without code changes.

Document workflow (contributor). Environment-based remapping of document state transitions across deployments.

Performance. Non-blocking index creation on production tables with millions of rows. Eliminated N+1 queries in hot endpoints through ORM annotations. Rewrote a cleanup query on a 280M-row table that was hitting statement timeout — replaced the query plan from nested loop to hash anti-join.

2020–2021

Logistics algorithms and visualization for large-scale geospatial data (vehicle tracks, nested territories, graph overlays) at regional scale.

2018–2019

Hardware-software complex for automating municipal yard territory inventory. Still serving original clients.

Teaching

Teaching assistant at MUCTR (2020), MIPT (2022). Code reviewer and mentor at Yandex.Practicum, backend development track (2023–2024).

Education

MSc in Computer Science — Moscow Institute of Physics and Technology (MIPT), 2022
Thesis: visualization algorithms for large-scale route data on maps.

BSc in Software Engineering — MIREA, 2020

Cisco CCNA (2019). Coursera: Practical Reinforcement Learning (HSE, 2021), Big Data Essentials: HDFS, MapReduce and Spark RDD (Yandex, 2020).

Projects

Radio Bargain Bot

Price-intelligence system for the secondary ham radio market. Aggregates listings from 98 sources across 50+ countries, scores each deal using Bayesian bootstrap benchmarks, publishes verified bargains to 12 regional Telegram channels. r2bpw.works/radio_bargain_bot_v2 · GitHub

More Fun to Receive — Custom SDR Receiver on FPGA

Custom SDR receiver firmware for Xilinx Spartan-6 (OSA103 Mini), written from scratch in Verilog: NCO, hardware-multiplier mixer, decimating CIC filter.

IR Bridge — Ericsson MC218 ↔ Flipper Zero ↔ LLM

Bidirectional infrared bridge from a 1999 Ericsson MC218 palmtop to a large language model. Flipper Zero decodes IrDA and relays via ESP32 to an LLM API. No cables, no IrDA stack. r2bpw.works/psion-flipper-bridge · GitHub

30m QRSS Beacons

Catalogue of identified QRSS beacons on the 30-metre band, built on a distributed network of KiwiSDR receivers worldwide. r2bpw.works/qrss